

Workshop Worksheet

Writing & Formatting a Research Article

Running example: **Incoming Tourist Trends in Thailand, 2017–2024**

Follow these steps in order. Each step matches a module and its **[LAB]** on the slides. By the end you will have a real LaTeX draft of a short tourism paper and a matching slide. Tick each box as you finish.

1. The dataset

Everyone uses the same file: `tourist-data.xlsx` (in the `WritingManuscript` folder). It has two sheets.

Sheet	Columns	Years
Tourist-Number-Data	Region, Nationality, Year, BE, Number	2017–2024
Tourist-Expense-Data	Region, Nationality, Year, BE, Total/Medical Expense, Ratio	2017–2023

The story to tell. Arrivals peaked at **39.9M in 2019**, collapsed to **0.51M in 2021** during COVID-19, then recovered to **35.5M by 2024**. Top 2024 source markets: China, Malaysia, India, South Korea. Your paper documents this collapse-and-recovery.

Sample abstract (use this in Step 2)

We examine inbound tourism to Thailand from 2017 to 2024 using official arrival and expenditure statistics. Arrivals rose from 35.6 million in 2017 to a pre-pandemic peak of 39.9 million in 2019, fell sharply to 0.51 million in 2021 amid COVID-19 border measures, and recovered to 35.5 million by 2024. We characterise the recovery across source regions and quantify the shifting composition of top markets. The findings inform tourism-demand forecasting and policy for resilient recovery.

2. Find a good venue

(Module 1 · 20 min)

- ☐ Open a Journal Finder (Elsevier / Springer / Taylor & Francis) **or** WikiCFP.
- ☐ Paste the working title and the sample abstract above.
- ☐ Choose **3 candidate venues** (tourism / hospitality / Asia-Pacific).
- ☐ Fill the table below from each venue's "About / For Authors" page.

Venue	Scope fit (1–5)	Time to 1st decision	Next deadline	Indexed?
.....				
.....				
.....				

Pick one to write toward: _____. Note its template name (e.g. `elsarticle`, `interact`).

3. Set up the LaTeX project on Overleaf

(Module 2 · 30 min)

- ☐ Create a free Overleaf account; *New Project* → start from your venue's template.
- ☐ Replace the title with “**Incoming Tourist Trends in Thailand, 2017–2024**” and add your name.
- ☐ Paste the sample abstract into the abstract field.
- ☐ Add four sections and two placeholders:

```
\section{Introduction}
\section{Data}
\section{Results}
% Fig.~\ref{fig:arr} and Table~\ref{tab:arrivals} go here (Steps 5--6)
\section{Discussion}
```

- ☐ Click **Recompile**; fix any red errors in the log until you get a clean PDF.

4. Literature review → reference database

(Module 4 · 25 min)

Search & discovery tools

- ☐ **Google Scholar** — search "Thailand inbound tourism", "COVID-19 tourism recovery", "tourism demand forecasting". Use “Cited by” to snowball; *Cite* → **BibTeX** to export single items; set an *Alert*.
- ☐ **ResearchRabbit** (free) — add a seed paper, explore the recommendation / citation network, sync to Zotero.

AI tools — use the free version (any account)

- ☐ **Consensus** — ask “*Has Thai inbound tourism recovered after COVID-19?*”; get an evidence-based answer with real citations.
- ☐ **Connected Papers** — enter a seed paper; read the visual graph to find foundational vs. derivative work.
- ☐ **SciSpace** — “chat with PDF” to extract each paper’s method, data, and results into a comparison table.

Note: Chula AIX (aix.car.chula.ac.th) provides paid tiers of these tools, usable at the Central Library on-site. For this workshop, the free versions are enough.

Draft the review with NotebookLM

- ☐ Upload your 8–10 PDFs as *sources* in NotebookLM.
- ☐ Prompt: "For each source, list the research question, method, data, and main finding." → paste the keys into your notes.
- ☐ Prompt: "Group these sources into themes and draft a literature-review outline." Verify every quote against the source.

Get your BibTeX (Zotero is optional)

- ☐ **Quickest:** in Google Scholar click *Cite* → *BibTeX* and paste into `references.bib`. Same from most DOI / publisher pages.
- ☐ Or ask **NotebookLM** / **SciSpace** for the BibTeX of each source — then verify authors, year, DOI.
- ☐ Set readable cite keys (e.g. `chen2022recovery`, `unwto2024barometer`); collect **8–10** entries.
- ☐ *Optional:* use Zotero (+ Better BibTeX) to manage many papers, then export the collection to BibTeX; upload `references.bib` to Overleaf.

AI ethics — before you go further

You are the author and are accountable; AI is never an author. **Verify every AI-suggested citation** against the real paper (chatbots fabricate references). Do not upload confidential or unpublished third-party material to public tools. Never let AI generate or alter data, results, or images. **Disclose** AI use per your venue: Elsevier wants a declaration in its own section before the references (no AI image generation); Taylor & Francis wants the tool name + version, how, and why in the Methods or Acknowledgements. **Always read your target venue's "Guide for Authors"**.

5. Citations with BibTeX

(Module 5 · 25 min)

Add to your preamble and end (classic BibTeX shown; use BibLaTeX if your template requires it):

```
\bibliographystyle{apalike}      % match your venue's required style
...
Arrivals rebounded strongly by 2024 \cite{chen2022recovery}.
...
\bibliography{references}         % at the end, before \end{document}
```

- ☐ Cite at least **4** references across Introduction and Discussion.
- ☐ Recompile (Overleaf runs bibtex automatically) until there are **no** “undefined citation” warnings.
- ☐ Confirm the reference list appears and is formatted in the venue's style.

6. AI: turn the Excel results into a LaTeX table

(Module 6 · 25 min)

- ☐ In `tourist-data.xlsx`, make a pivot of **total arrivals by year (2017–2024)** (sum of Number grouped by Year).
- ☐ Copy the pivot as CSV text, then give your AI assistant this prompt:

Copy-paste prompt for your AI assistant

I use the `elsarticle` class with `booktabs`. Convert this CSV into a LaTeX table, with a caption “Thai tourist arrivals by year” and label `tab:arrivals`. Right-align the numbers, give compilable code, and list any packages needed.

```
Year,Arrivals
2017,35591978
2018,38178194
...
2024,35545714
```

- ☐ Paste the AI output into Overleaf; compile.

- ☐ **Verify every number** against the answer key below; reference the table in the text with Table \ref{tab:arrivals}

Answer key (verify against this):

Year	Arrivals	Year	Arrivals
2017	35,591,978	2021	510,767
2018	38,178,194	2022	11,065,226
2019	39,916,251	2023	28,150,016
2020	6,725,193	2024	35,545,714

7. AI: draw a chart and import the graphic

(Module 7 · 25 min)

- ☐ Ask your AI assistant for a plotting script (prompt below), then run it. **No Python installed?** Run it free in Google Colab (*New notebook* → upload the xlsx → Run); pandas/matplotlib are pre-installed.

Copy-paste prompt for your AI assistant

Write a Python script that reads tourist-data.xlsx sheet Tourist-Number-Data, sums Number by Year, and plots a line chart of arrivals (in millions) for 2017–2024 with labeled axes and markers. Save it as a vector PDF arrivals.pdf. No chart title.

Expected starter script:

```
import pandas as pd, matplotlib.pyplot as plt
df = pd.read_excel("tourist-data.xlsx", sheet_name="Tourist-Number-Data")
s = df.groupby("Year")["Number"].sum() / 1e6
plt.figure(figsize=(5,3)); plt.plot(s.index, s.values, marker="o")
plt.ylabel("Arrivals (millions)"); plt.xlabel("Year")
plt.tight_layout(); plt.savefig("arrivals.pdf")
```

- ☐ Check the curve: peak in 2019, trough in 2021, recovery by 2024.
- ☐ Upload arrivals.pdf to Overleaf and import it:

```
\begin{figure}[t]\centering
\includegraphics[width=.7\linewidth]{arrivals.pdf}
\caption{Thai tourist arrivals, 2017--2024.}\label{fig:arr}
\end{figure}
```

- ☐ Reference it in the text: “As Fig. \ref{fig:arr} shows, arrivals collapsed in 2020–21 and recovered by 2024.”
- ☐ Recompile to a clean PDF.
- ☐ **Bonus:** ask AI for a TikZ or Mermaid diagram of your workflow (Excel → Python → Figure → LaTeX), or a bar chart of the top-10 source markets in 2024.

8. Equations: typeset the Gaussian analysis

(Module 8 · 20 min)

- ☐ In Colab, fit a normal distribution to $\ln(\text{arrivals})$ for 2024 and read off μ, σ .

Copy-paste prompt for your AI assistant

In tourist-data.xlsx (sheet Tourist-Number-Data), take the 2024 rows with arrivals > 0, compute $x = \ln(\text{Number})$, report its mean and standard deviation, and plot a density histogram of x with the fitted normal curve overlaid. Save it as gaussian_fit.pdf.

- ☐ You should get about $\mu \approx 11.1$ and $\sigma \approx 2.0$ (median $e^\mu \approx 67,000$ arrivals per market).
- ☐ Add a Methods section with the Gaussian PDF — LaTeX renders math far better than a word processor:

```
\usepackage{amsmath} % in the preamble
...
\begin{equation}
f(x)=\frac{1}{\sigma\sqrt{2\pi}}
\exp\{-\frac{(x-\mu)^2}{2\sigma^2}\}
\label{eq:gauss}
\end{equation}
Log-arrivals are well described by Eq.~\eqref{eq:gauss}.
```

- ☐ Include `gaussian_fit.pdf` as a figure, refer to it with Eq. `\eqref{eq:gauss}`, and recompile.

9. From paper to slides: Beamer

(Module 9 · 20 min)

- ☐ Start a new *Beamer* project on Overleaf (`\documentclass{beamer}`).
- ☐ Reuse your manuscript: copy the Gaussian equation and `gaussian_fit.pdf` into one frame.

```
\documentclass{beamer}
\usepackage{amsmath}
\begin{document}
\begin{frame}{Log-arrivals are Gaussian}
\begin{equation*}
f(x)=\frac{1}{\sigma\sqrt{2\pi}}
\exp\{-\frac{(x-\mu)^2}{2\sigma^2}\}
\end{equation*}
\includegraphics[width=.6\linewidth]{gaussian_fit.pdf}
\end{frame}
\end{document}
```

- ☐ Add a second frame with your arrivals table or chart — copy-paste from the paper, nothing redrawn.
- ☐ Compile. The equation and figure are *identical* to the manuscript: one source of truth.

10. Finish line: what you should have

- ☐ A 3-row venue comparison table and one chosen target venue.
- ☐ An Overleaf project on a real template: title, authors, abstract, sections.
- ☐ `references.bib` with 8–10 references, 4+ cited in the text.
- ☐ Table `tab:arrivals` (arrivals by year), numbers verified.
- ☐ Figure `arrivals.pdf` imported and cross-referenced.
- ☐ A Methods section with the Gaussian equation (`eq:gauss`) and `gaussian_fit.pdf`.
- ☐ A Beamer slide that reuses the same equation and figure.
- ☐ A compiled PDF with **no** errors or undefined-reference warnings.

Remember: AI drafts code, tables, and figures fast — but **you** verify every number against `tourist-data.xlsx` and **you** are accountable for the result. Disclose AI use per your venue's policy.